

PETER A. SAYEGH

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EDUCATION

Columbia University

New York, NY

MS in Electrical Engineering – Nikola Tesla Scholarship, GPA: 3.79/4.0

Expected Dec 2026

Coursework: Digital Signal Processing, Adv. Analog Integrated Circuits, Power Electronics, MOS Transistors, Analog-Digital Interfaces in VLSI, Embedded AI, Microwave Circuit Design, Motor Drive Systems, High-Frequency Magnetics Design

Activities: Lead Department Representative for Electrical Engineering within the Engineering Student Graduate Council (EGSC)

École Polytechnique

Palaiseau, FR

BS in Mathematics and Physics, GPA: 3.75/4.0

Jun 2025

Coursework: Convex Optimization, Stochastic Processes, Electrodynamics, Adv. Quantum Physics, Thermodynamics, Statistical Physics, Condensed Matter, Algorithms, Numerical Analysis, Web Programming, Object-Oriented Programming, Machine Learning

EXPERIENCE

Columbia Engineering

New York, NY

Course Assistant

Sep 2026 – Present

- Supported Prof. Guido Jajamovich and his students in Digital Signal Processing by holding weekly office hours and review sessions and grading homework, assignments, and projects

Illumination Diagnostics

New York, NY

Electrical Engineering Intern

Jun - Aug 2026

- Developed a 31 Hz heart rate extraction pipeline for a 12-channel optical wearable, achieving 1-3 bpm clinical accuracy
- Prototyped PPG and temperature acquisition systems using ESP32 microcontrollers with synchronized respiration belt recording
- Assembled a 940 nm IR LED illumination array on perfboard (~14 mA/LED) to provide stable background illumination, reducing ambient light interference in pupil tracking
- Designed a 2-axis pan/tilt laser targeting system (dual servos, Arduino + Waveshare driver), achieving $<0.1^\circ$ angular resolution for programmable wall-target positioning

ELECTRICAL ENGINEERING PROJECTS

Columbia University

New York, NY

Fully Differential Switched-Capacitor Amplifier

Feb – May 2026

- Designed folded-cascode OTA with Common-mode Feedback in $0.18\ \mu\text{m}$ CMOS, achieving 56.6 dB gain, 71° phase margin, and 333 dB CMRR at 486 μW
- Derived settling and slew-rate constraints to meet 0.09% settling error within 28 ns for a 7-bit 16 MHz pipelined ADC

Columbia University

New York, NY

SignSpeak: ASL Recognition on Microcontroller

Feb – May 2026

- Deployed 16K-parameter depthwise-separable CNN on ESP32S3 via TFLite Micro INT8 quantization, achieving 97.4% accuracy in 67.4 KB flash
- Built a full on-device inference pipeline from camera capture to OLED display, <110 ms latency with no cloud dependency

Columbia University

New York, NY

250 MS/s 8-bit Current-Steering DAC

Feb - May 2026

- Designed a 3+5 segmented current-steering DAC in $0.18\ \mu\text{m}$ CMOS with thermometer MSBs and per-cell trim DACs, achieving SFDR >66 dB and DNL $<2 \times 10^{-4}$ LSB across all process corners
- Optimized cascode unit cell sizing to meet 1% mismatch budget, delivering $\geq 1.53\ V_{pp,diff}$ into $100\ \Omega$ at 250 MS/s within 27.6 mW

École Polytechnique - Center for Theoretical Physics

Palaiseau, FR

Dynamic Stability and Levitation Control in Magnetic Levitation (Maglev) Systems

Feb 2024 - Jul 2026

- Designed and constructed a scaled Maglev train prototype using magnetic field analysis, achieving 96% theoretical validation
- Extrapolated results to a real Electromagnetic Suspension engine scale, predicting capacity within 2% of Shanghai Maglev
- Submitted a publication to the European Journal of Physics, which is currently provisionally accepted

TECHNICAL SKILLS

CAD Tools: Cadence Spectre, PLECS, LTSpice, Keysight ADS, Xschem, ngspice, Magic VLSI, KLayout, OpenROAD

Programming Languages: Python, C++, Arduino, R, MATLAB, HTML, CSS, JavaScript, QML

Data Science and Machine Learning: TensorFlow, Keras, JAX/Optax, scikit-learn, NumPy, SciPy, pandas, statsmodels, DSPy

Scientific Computing: QuTiP, Scikit-HEP, PySpice

Languages: French (Native), Arabic (Native), German (Advanced), Spanish (Intermediate)